

Probabilistic Inference

Probabilities are the mathematical formalisation for the concept of *uncertainty*. Many numerical tasks involve quantities taking values x in the real vector space $\mathbb{R}^D$ for $D \in \mathbb{N}$. So consider a continuous variable $X \in \mathbb{R}$, and assume that its value x is not known precisely. Instead, it is only known that the *probability* for the value of X to fall into the set $U \subset \mathbb{R}$ is $P_X(U)$. If P_X is sufficiently regular, one can define the probability density function $p(x)$ as the Radon–Nikodym derivative of P_X with respect to the Lebesgue measure. That is, the function $p(x)$ with the property $P_X(U) = \int_U p(x) dx$ for all measurable sets $U \subset \mathbb{R}$. There are two basic rules for the manipulation of probabilities: If two variables $x, y \in \mathbb{R}$ are assigned the density function $p(x, y)$, then the *marginal* distribution is given by the *sum rule*

$$p(y) = \int p(x, y) dx,$$

and the *conditional* distribution $p(x | y)$ for x given that $Y = y$ is provided implicitly by the *product rule*

$$p(x | y)p(y) = p(x, y), \quad (2.1)$$

whose terms are depicted in Figure 2.1. The corollary of these two rules is *Bayes' theorem*, which describes how *prior knowledge*, combined with *data* generated according to the conditional density $p(y | x)$, gives rise to the *posterior* distribution on x :

$$\underbrace{p(x | y)}_{\text{posterior}} = \frac{\overbrace{p(y | x)}^{\text{likelihood}} \overbrace{p(x)}^{\text{prior}}}{\underbrace{\int p(y | x)p(x) dx}_{\text{evidence}}}.$$

When interpreted as a function of x , the conditional distribution $p(y | x)$ is called the *likelihood* (note it is not a probability density

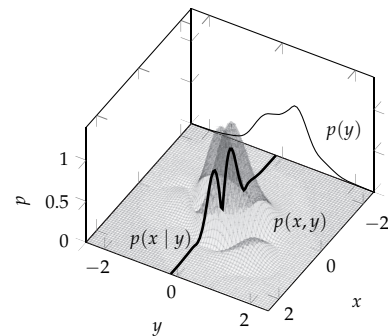


Figure 2.1: Conceptual sketch of a joint probability distribution $p(x, y)$ over two variables x, y , with marginal $p(y)$ and a conditional $p(x | y)$.

function of x , but of y). Computing the posterior amounts to *inference* on x from y .

Although not without competition, probability theory is widely accepted as the formal framework for reasoning with imperfect knowledge (uncertainty) in the natural sciences,¹ statistics,² and computer science.³ This text accepts the general philosophical and mathematical arguments made in these texts and asks, more practically, how to use probabilistic formulations in computation.

¹ Jaynes and Bretthorst (2003)

² Cox (1946); Le Cam (1973); Ibragimov and Has'minskii (1981)

³ Pearl (1988); Hutter (2010).